Question 1

(a) True

Let (x_n) be a convergent sequence in X, converging to $x \in X$. Since X is finite, (x_n) is eventually constant (Hand-in 1). That is, there exists some $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n = x$. That means the sequence $(f(x_n))$ is eventually constant, as for all $n \geq N, f(x_n) = f(x)$. So $f(x_n) \to f(x)$ as $n \to \infty$ and thus f is continuous

(b) False.

Consider metric spaces $(\mathbb{R}, d_E)$ and $(\mathbb{R}, d_D)$, and denote $\mathbb{R}_E$ as the set from the euclidean metric space and $\mathbb{R}_D$ for the discrete space. For $f : \mathbb{R}_D \to \mathbb{R}_E$ defined by $f(x) = x$, clearly f is bijective as $\mathbb{R}_D = \mathbb{R}_E = \mathbb{R}$. Since if (x_n) in $\mathbb{R}_D$ converges to $x \in \mathbb{R}_D$ with respect to d_D, the sequence (x_n) must be eventually constant and thus the sequence $(f(x_n))$ must converge to $f(x)$ with respect to d_E. So f is continuous. Now we use Proposition 6.32 to prove that f^{-1} is not continuous. The set $(0, 1)$ is closed in $\mathbb{R}_D$ (every set is a closed set), but $f((0, 1)) = (0, 1)$ is not closed in $\mathbb{R}_E$ (for example, sequence $(\frac{1}{n})$ converges to $0 \notin (0, 1)$) so f is not closed and thus f^{-1} is not continuous

(c) False.

Let $X = [0, 1]$ and $Y = \mathbb{R}$, both equipped with the normal euclidean metric and define $f : [0, 1] \to \mathbb{R}$ by $f(x) = x$. Then $d_Y(f(x_1), f(x_2)) = d_Y(x_1, x_2) = d_E(x_1, x_2) = d_X(x_1, x_2)$ for all $x_1, x_2 \in [0, 1]$ making f an isometry but is clearly not surjective

(d) False.

Consider $X = Y = [-1, 1]$, both equipped with the euclidean metric. Then for $f : [-1, 1] \to [-1, 1]$ defined by $f(x) = x^2$, we get that for all $x, y \in [-1, 1]$, $d_E(f(x), f(y)) = |x^2 - y^2| = |x - y||x + y| \leq |x - y|(|x| + |y|) \leq 2|x - y|$ since $|x|, |y| \leq 1$. So f is Libschitz-continuous, but is not injective since $f(-1) = f(1) = 1$

(e) True.

Let $\epsilon > 0$. Since f is Lipschitz-continuous, for all $x, y \in X$, $d_Y(f(x), f(y)) \leq kd_X(x, y)$, for some constant $k > 0$. Let $\delta = \frac{\epsilon}{k}$. Now let $x, y \in X$ and $d_X(x, y) < \frac{\epsilon}{k}$. Then $d_Y(f(x), f(y)) \leq kd_X(x, y) < k \cdot \frac{\epsilon}{k} = \epsilon$ as required and thus f is uniformly continuous

Question 2

Let $X = \{x, y\}$. For $\mathcal{T}$ to be a topology, $\mathcal{T}$ is a family of subsets of X such that:
(T1): $X \in \mathcal{T}$ and $\emptyset \in \mathcal{T}$
(T2): Any finite intersection of sets in $\mathcal{T}$ is still in $\mathcal{T}$.
(T3): Arbritrary unions of sets in $\mathcal{T}$ is still in $\mathcal{T}$.
Note $\mathcal{P}(X) = \{\emptyset, \{x\}, \{y\}, \{x, y\}\}$. All possible families of subsets containing $\emptyset$ and X are:
$\mathcal{T}_1 = \{\emptyset, \{x, y\}\}$
$\mathcal{T}_2 = \{\emptyset, \{x\}, \{x, y\}\}$
$\mathcal{T}_3 = \{\emptyset, \{y\}, \{x, y\}\}$
$\mathcal{T}_4 = \{\emptyset, \{x\}, \{y\}, \{x, y\}\}$
All these sets satisfy (T2) and (T3) so this is a list of all possible topologies of X
Notice that $\mathcal{T}_1$ and $\mathcal{T}_4$ are not homeomorphic to any other topology, since any other topology has a different size and and thus a different finite number of open sets (so any bijection f would map n open sets to n open sets). Now $\mathcal{T}_2$ is homeomorphic to $\mathcal{T}_3$ using $f : X \to X$ with $x \mapsto y$ and $y \mapsto x$. This is then the only homeomorphic topological space